Vectors and Co-Vectors

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April 2026

1 What is a vector really?

let's say you were running up a hill. You started from the bottom (Point A) and ended at somewhere in the middle of the hill (point B). Let's see this on a diagram.

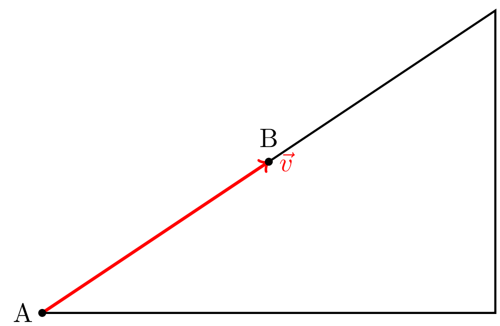

Figure 1: Vector $\vec{v}$ from point A to B.

It is important to note that the arrow does not represent the exact path taken by the runner. The runner may have taken a curved or uneven route.

The vector only captures the overall change from A to B, ignoring the details of the journey.

A vector is often described as an object that has magnitude and direction. That is, it tells us how far something goes, and in what direction.

Now consider the red arrow in the diagram. It points from A to B, and it clearly has a certain length. So it seems to fit this description. But there is something subtle going on. The diagram does not tell us what we mean by directions such as left, right, up, or down in any precise way. It also does not tell us how to measure length. We have not said what counts as one unit of distance.

Does this mean the red object is not a vector? No.

The vector is still there. It represents the displacement from A to B. It tells us how to move from the starting point to the ending point. What is missing is not the vector itself, but a way of describing it using numbers.

If we decide how to measure distances, and if we agree on how to describe directions, then we can assign numbers to this vector. We could then say exactly how long it is, and describe its direction more precisely.

So the vector itself does not depend on these choices, but any numerical description of it does.

This raises an important question: how do we actually describe a vector using numbers?

To do that, we will need to introduce a way of measuring lengths and describing directions more precisely.

2 Describing direction and length

In the previous section, we saw that a vector represents a displacement from one point to another. It tells us how to move, but we have not yet said how to describe that movement using numbers.

If we want to be more precise, we need two things:

1. a way to measure distance,
2. a way to describe direction.

Let us start with distance. We shall introduce a grid to our previous diagram.

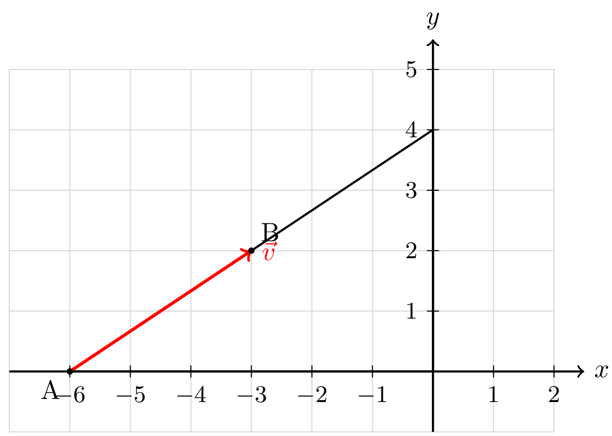

Figure 2: Vector $\vec{v}$ from point A to B, shown on a grid with axes for measurement.

Now that we have introduced a grid into the diagram, we can use it as a tool for measurement. Each square in the grid can be thought of as representing one unit of length.

With this in place, we no longer need to guess how long the vector is. We can estimate its length by counting how many units it covers.

Now we turn to direction. The grid also helps us compare directions in a consistent way. The horizontal lines of the grid give us a natural notion of left and right, while the vertical lines give us a notion of up and down.

These directions do not come from the vector itself; they are choices that we make in order to describe motion more precisely.

Once these reference directions are fixed, we can describe any vector by saying how much it moves in each of these two directions.

In other words, instead of thinking of the vector as a single arrow, we can think of it as being built from two simpler motions: one along the horizontal direction, and one along the vertical direction.

Using the grid, we can now read off these motions directly. We can see how far the vector moves horizontally, and how far it moves vertically.

This allows us to assign numbers to the vector. These numbers tell us how much of each basic direction is needed to reconstruct the original arrow.

Thus, while the vector itself is a geometric object, our description of it depends on the choices we make for measuring distance and describing direction.

From the grid, we can read off that the vector moves 3 units in the horizontal direction and 2 units in the vertical direction.

This raises an important question: why were we able to describe the vector in this way?

The reason is that we have chosen two special directions in the plane: one horizontal and one vertical. These directions act as our fundamental building blocks. By combining them, we are able to describe any displacement in the plane.

3 Basis and Vector Spaces

The two directions we chose, horizontal and vertical, play a special role. They allow us to describe every vector in the plane by breaking it into simpler pieces.
We call such a collection of directions a **basis**.

Very simply, a **basis** is a set of directions with the property that every vector can be built by combining them in suitable amounts.

A precise definition of a basis:
So far, we have been using the horizontal and vertical directions as our basic building blocks. We saw that by combining these directions, we were able to construct the vector $\vec{v}$.
We now want to make this idea precise.
To do this, we introduce the notion of a **vector space**. A vector space is a collection of objects, called vectors, together with two operations:

1. vector addition, which combines two vectors to form another vector, and
2. scalar multiplication, which scales a vector by a number.

These operations satisfy certain properties (such as associativity, distributivity, and the existence of a zero vector), which ensure that vectors behave in a consistent algebraic way.

Within a vector space, we are interested in describing vectors in terms of simpler building blocks.

Linear combinations

Given vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, we can form a new vector $\vec{w}$ by taking a linear combination:

$$\vec{w} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n,$$

where $a_1, a_2, \ldots, a_n$ are real numbers.
This represents combining the vectors by scaling each one and then adding the results.

Spanning

A collection of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is said to **span** the vector space if every vector in the space can be written as a linear combination of them.
In other words, these vectors are sufficient to build every vector in the space.

Linear independence

However, not all spanning sets are equally useful. We would like our building blocks to be as efficient as possible.
A set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is said to be **linearly independent** if the only way to write

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_n\mathbf{v}_n = \mathbf{0}$$

is by taking

$$a_1 = a_2 = \cdots = a_n = 0.$$

This means that none of the vectors can be built from the others. Each one contributes something genuinely new.

Definition of a basis

We are now ready to define a basis.
A set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is called a **basis** of a vector space if:

1. it spans the space, and
2. it is linearly independent.

This means that a basis is a minimal set of building blocks from which every vector in the space can be constructed.

Connection to our example

In our diagram, the horizontal direction $\mathbf{e}_1$ and the vertical direction $\mathbf{e}_2$ form a basis of the plane.
Every vector can be written uniquely as a combination of these two directions. For example,

$$\vec{v} = 3\mathbf{e}_1 + 2\mathbf{e}_2.$$

The numbers $(3, 2)$ are called the **components** of the vector with respect to this basis. These components depend on the choice of basis, even though the vector itself does not.

It is important to note that the choice of basis is not unique. We could have chosen a different pair of directions, and we would still be able to describe every vector in the plane. However, the numerical components of the vector would then change accordingly.

4 The hill example revisited

We now return to our hill example and reinterpret it using the language of vector spaces and bases.

What is the vector space here?

In this example, we are working in the plane of the diagram. The vectors we are considering are all possible displacements that can be drawn on this plane.
That is, starting from any point, we can imagine moving in any direction and by any amount. All such displacements form our vector space.
This space is two-dimensional, since two independent directions are sufficient to describe any motion within it.

What is the basis in this example?

When we introduced the grid, we implicitly chose two special directions:

- the horizontal direction (along the x-axis), and
- the vertical direction (along the y-axis).

We denote these directions by $\mathbf{e}_1$ and $\mathbf{e}_2$, respectively.
These two vectors form a basis of the plane. They are linearly independent, and together they span all possible directions in the diagram.

Writing the vector in this basis

Now consider the vector $\vec{v}$ from point A to point B.
From the grid, we can see that:

- the vector moves 3 units in the horizontal direction, and
- the vector moves 2 units in the vertical direction.

This means that $\vec{v}$ can be written as a linear combination of the basis vectors:

$$\vec{v} = 3\mathbf{e}_1 + 2\mathbf{e}_2.$$

This expression tells us exactly how to reconstruct the vector: we take 3 copies of the horizontal direction and 2 copies of the vertical direction, and add them together.

A deeper observation

It is important to note that the vector $\vec{v}$ itself is not the pair of numbers $(3, 2)$.
Rather, $(3, 2)$ is the description of the vector with respect to the chosen basis $\{\mathbf{e}_1, \mathbf{e}_2\}$.
If we had chosen a different pair of directions as our basis, the same vector would be described by a different pair of numbers.

Thus, the vector is a geometric object, while its components depend on the basis used to describe it.

5 Changing the basis

So far, we have described vectors using the horizontal and vertical directions as our basis. This allowed us to write the vector $\vec{v}$ as

$$\vec{v} = 3\mathbf{e}_1 + 2\mathbf{e}_2.$$

However, there is nothing special about these particular directions. We are free to choose a different set of directions, as long as they are linearly independent and span the space.

A new choice of basis

Suppose instead of choosing purely horizontal and vertical directions, we choose directions that are tilted along the hill.
For example, let us define:

- $\mathbf{u}_1$ as a direction pointing up along the slope of the hill,
- $\mathbf{u}_2$ as another direction that is not parallel to $\mathbf{u}_1$.

As long as these two directions are not multiples of each other, they will still form a basis of the plane.

Describing the same vector

Now we try to describe the same vector $\vec{v}$ using this new basis $\{\mathbf{u}_1, \mathbf{u}_2\}$.
Instead of moving purely horizontally and then vertically, we now think of the vector as being built from motions along these new directions.
This means we can write

$$\vec{v} = a\,\mathbf{u}_1 + b\,\mathbf{u}_2$$

for some numbers a and b.

What has changed?

The vector $\vec{v}$ has not changed. It still represents the same displacement from A to B.
What has changed is only the way we describe it.
Previously, we said the vector was made of 3 units in the horizontal direction and 2 units in the vertical direction.
Now, we describe it in terms of how much it moves along $\mathbf{u}_1$ and $\mathbf{u}_2$.
So the numbers $(3, 2)$ will generally be replaced by a different pair of numbers (a, b).

A key insight

This leads to an important conclusion:

- The vector itself is independent of the basis.
- The components of the vector depend on the basis.

Thus, changing the basis changes the numerical description of the vector, but not the geometric object itself.

A concrete example

Let us choose a new basis:
$\vec{u}1$ and $\vec{u}2$

We now try to express $\vec{v} = 3\mathbf{e}_1 + 2\mathbf{e}_2$ in terms of $\mathbf{u}_1$ and $\mathbf{u}_2$.

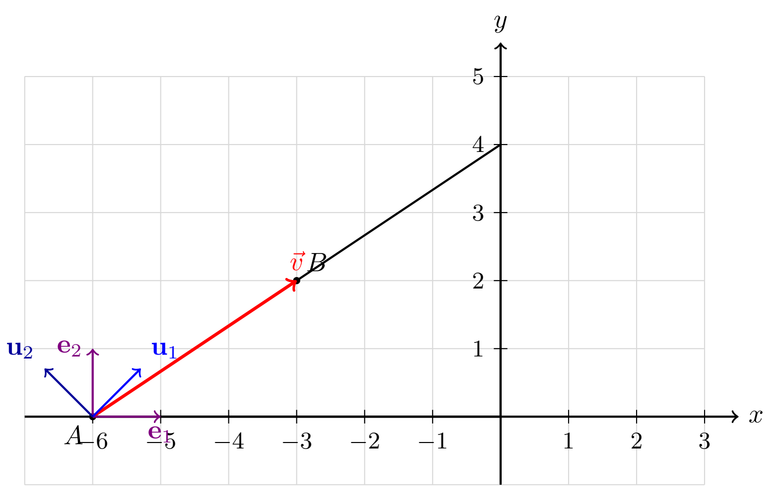

Figure 3: The vector $\vec{v}$ together with the standard basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ and the rotated basis $\{\mathbf{u}_1, \mathbf{u}_2\}$ obtained by rotating the standard basis anticlockwise by 45°.

Since the new basis vectors are obtained by rotating the standard basis by 45°, we have

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(\mathbf{e}_1 + \mathbf{e}_2), \qquad \mathbf{u}_2 = \frac{1}{\sqrt{2}}(-\mathbf{e}_1 + \mathbf{e}_2).$$

We now solve for $\mathbf{e}_1$ and $\mathbf{e}_2$ in terms of $\mathbf{u}_1$ and $\mathbf{u}_2$.
Adding the two equations:

$$\mathbf{u}_1 + \mathbf{u}_2 = \frac{1}{\sqrt{2}}(2\mathbf{e}_2),$$

so

$$\mathbf{e}_2 = \frac{1}{\sqrt{2}}(\mathbf{u}_1 + \mathbf{u}_2).$$

Subtracting the second equation from the first:

$$\mathbf{u}_1 - \mathbf{u}_2 = \frac{1}{\sqrt{2}}(2\mathbf{e}_1),$$

so

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}}(\mathbf{u}_1 - \mathbf{u}_2).$$

Now substitute into $\vec{v} = 3\mathbf{e}_1 + 2\mathbf{e}_2$:

$$\vec{v} = 3\left(\frac{1}{\sqrt{2}}(\mathbf{u}_1 - \mathbf{u}_2)\right) + 2\left(\frac{1}{\sqrt{2}}(\mathbf{u}_1 + \mathbf{u}_2)\right).$$

Factor out $\frac{1}{\sqrt{2}}$:

$$\vec{v} = \frac{1}{\sqrt{2}}\left[3(\mathbf{u}_1 - \mathbf{u}_2) + 2(\mathbf{u}_1 + \mathbf{u}_2)\right].$$

Expand:

$$\vec{v} = \frac{1}{\sqrt{2}}\left(3\mathbf{u}_1 - 3\mathbf{u}_2 + 2\mathbf{u}_1 + 2\mathbf{u}_2\right).$$

Simplify:

$$\vec{v} = \frac{1}{\sqrt{2}}\left(5\mathbf{u}_1 - \mathbf{u}_2\right).$$

Thus, in the rotated basis, the vector is

$$\vec{v} = \frac{5}{\sqrt{2}}\,\mathbf{u}_1 - \frac{1}{\sqrt{2}}\,\mathbf{u}_2.$$

This shows that the same vector can have different numerical descriptions depending on the basis used. The components are not intrinsic to the vector; they arise from our choice of basis.